An English to Spanish Translator

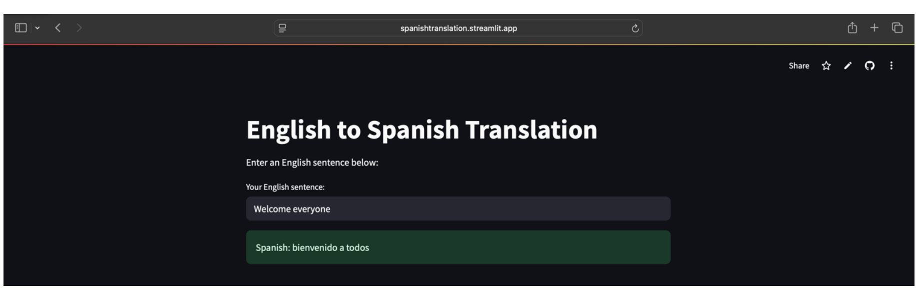
Introduction: This project presents a fully functional English-to-Spanish translator built using a Transformer-based sequence-to-sequence model. The architecture incorporates positional encoding, multi-head self-attention, and feedforward layers, offering efficient and scalable performance for language translation. The model was trained on bilingual sentence pairs and achieved approximately 70% validation accuracy over the course of 10 training epochs, demonstrating effective learning of cross-linguistic mappings.

Model Design and Training: The model was implemented using TensorFlow and Keras and trained on parallel English-Spanish sentence pairs. It was configured with 4 Transformer layers, 128-dimensional embeddings, 8 attention heads, and a feedforward dimension of 512.

Vectorization Preservation: A TextVectorization layer with a registered custom standardization function was used to normalize input sequences by lowercasing and removing punctuation. To ensure this function could be saved and reloaded along with the vectorizer model, it was annotated with @register_keras_serializable(). This decorator makes the function discoverable by Keras during model deserialization, ensuring that vectorization behavior remains consistent across training and inference. Both the source and target vectorization layers were saved as .keras files. This preserved the learned vocabulary and token-index mappings, which are essential for accurate input encoding and output decoding. Saving these layers prevents discrepancies that could arise from reinitializing or regenerating vocabularies post-training.

Inference Pipeline: The trained model weights were saved in .h5 format and loaded into a reconstructed Transformer model. A dummy batch was passed during reinitialization to build the model structure before applying the saved weights. This setup enables the model to be used repeatedly without retraining, ensuring reproducibility and modularity.

User Interaction: The submission includes both a Jupyter notebook and a live Streamlit web app for real-time English-to-Spanish translation. In the notebook version, users enter an English sentence, which is preprocessed, passed through the trained Transformer model, and decoded using greedy decoding to produce the Spanish output. The Streamlit app replicates this functionality in a web interface, allowing users to interact with the model through a browser. Hosted on Streamlit Cloud and linked to GitHub, it ensures ease of access and simulates a production-ready experience. Link: https://spanishtranslation.streamlit.app



Conclusion: This project offers a robust and deployable translation system using Transformer-based deep learning. Its modular design, consistent vectorization, and interactive inference showcase a strong grasp of model architecture and real-world NLP deployment.

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